Options & Trading

Markets, Option Products, the Greeks, and Managing Risk

A one-hour introduction · formal enough to be precise, light enough to follow

What we will cover

Part I — Foundations

1. What trading is, and who trades
2. Stocks and the price process
3. Risk, return, and volatility

Part II — Options

4. Calls, puts, and payoffs
5. What an option is worth
6. Black–Scholes in one slide

Part III — The Greeks

7. Delta and hedging
8. Gamma, Theta, Vega
9. Implied volatility & the smile

Part IV — Strategies & Risk

10. Directional & volatility strategies
11. Managing a book with the Greeks

Goal: understand *why* options exist, *how* they are priced, and *how* they are used to shape and control risk.

Part I — What is trading?

Trading is the voluntary exchange of a financial asset for cash at an agreed price. A market is simply the place where buyers and sellers meet.

Key quantities

- **Bid** — best price a buyer offers
- **Ask** — best price a seller asks
- **Spread** $=$ Ask $-$ Bid
- **Liquidity** — how easily you trade without moving the price

Why people trade

- **Investing** — own a claim on future cash flows
- **Hedging** — reduce an existing risk
- **Speculation** — take a view on the price
- **Market-making** — earn the spread by providing liquidity

A trade has *two* sides. Your profit is someone's cost — pricing is about agreeing on a *fair* number.

Stocks: a share of a company

A stock (share) is a unit of ownership in a company. Owning n shares of a firm with N shares outstanding entitles you to a fraction n/N of its equity, its dividends, and its votes.

- **Price** S_t — what one share trades for at time t.
- **Market capitalisation** $= N \cdot S_t$ — the firm's equity value.
- **Dividend** — periodic cash paid to shareholders.
- **Total return** combines price change *and* dividends.

The price is set continuously by supply and demand. From a trader's point of view S_t is best modelled not as a fixed number but as a random process — we cannot predict tomorrow's price, only describe its distribution.

The price as a stochastic process

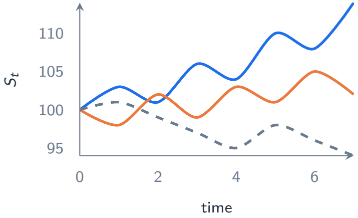
A standard model for a non-dividend stock is Geometric Brownian Motion (GBM):

$$\mathrm{d}S_t = \underbrace{\mu\, S_t\, \mathrm{d}t}_{\text{drift (trend)}} + \underbrace{\sigma\, S_t\, \mathrm{d}W_t}_{\text{diffusion (noise)}},$$

where W_t is Brownian motion, μ the expected growth rate and σ the volatility. Its solution is log-normal:

$$S_T = S_0 \exp\Big[\big(\mu - \tfrac{1}{2}\sigma^2\big)\, T + \sigma\sqrt{T}\, Z\Big], \qquad Z \sim \mathrm{N}(0,1).$$

- Returns are random; *log*-returns are (roughly) normal.
- μ tilts the average path; σ sets its width.
- More time $\Rightarrow$ more uncertainty ($\propto \sqrt{T}$).

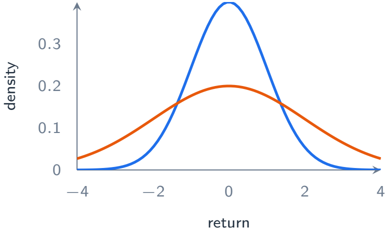


Risk, return, and volatility

Over a horizon, the (log) return is a random variable. We summarise it with its first two moments:

$$\text{return } R = \frac{S_T - S_0}{S_0}, \qquad \mu_R = \mathbb{E}[R] \text{ (reward)}, \qquad \sigma_R = \sqrt{\mathrm{Var}[R]} \text{ (risk)}.$$

- **Volatility** σ is the standard deviation of returns — the single most important number for options.
- It scales with the square root of time: $\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$.
- Higher $\sigma \Rightarrow$ wider outcomes $\Rightarrow$ *more valuable* optionality.



low-σ (blue) vs. high-σ (orange)

Risk is not the enemy — it is the *raw material* options are built from.

Part II — What is an option?

An option is a contract giving its holder the *right, but not the obligation*, to buy or sell an underlying asset at a fixed strike price K before or at expiry T.

Call option

- Right to **buy** at K.
- Bet that S will **rise**.
- Payoff at T: $\max(S_T - K, 0)$.

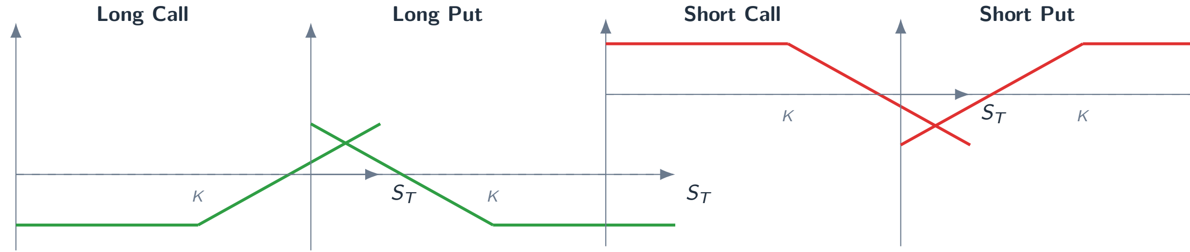
Put option

- Right to **sell** at K.
- Bet that S will **fall**.
- Payoff at T: $\max(K - S_T, 0)$.

- The buyer pays a premium up front for this right.
- **American** options exercise any time up to T; **European** only at T.
- Every contract has a **buyer** (long, limited loss) and a **seller/writer** (short, obligated).

Payoffs at expiry

The value at expiry depends only on where S_T lands relative to K.



- Solid line = payoff *net of premium*; the dashed line is break-even.
- Buyers have **capped loss** (the premium) and **convex** upside.
- Writers collect the premium but carry **open-ended** risk — the mirror image.

What is an option worth before expiry?

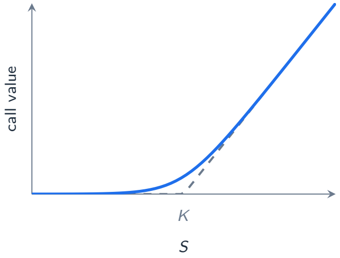
Before expiry, an option's premium splits into two parts:

$$\underbrace{C_t}_{\text{price}} = \underbrace{\max(S_t - K, 0)}_{\text{intrinsic value}} + \underbrace{\text{time value}}_{\geq 0,\ \text{from future uncertainty}}.$$

Moneyness (for a call):

- **ITM** — in the money, $S > K$
- **ATM** — at the money, $S \approx K$
- **OTM** — out of the money, $S < K$

Time value is *largest* at the money and *decays* to zero as $t \to T$ ("theta decay").



price (blue) sits above intrinsic (dashed)

Black–Scholes in one slide

Under GBM with constant volatility σ and risk-free rate r, the fair price of a *European call* is

$$C \;=\; S_0\,\mathrm{N}(d_1) \;-\; K\,e^{-rT}\,\mathrm{N}(d_2), \qquad d_{1,2} = \frac{\ln(S_0/K) + \left(r \pm \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

Where it comes from

- Build a portfolio of the option *hedged* with Δ shares so it is momentarily **risk-free**.
- A risk-free portfolio must earn the risk-free rate r — no free lunch (no arbitrage).
- That argument becomes a PDE whose solution is the formula above.

What drives the price

- $S_0 \uparrow \Rightarrow C \uparrow$
- $K \uparrow \Rightarrow C \downarrow$
- $\sigma \uparrow \Rightarrow C \uparrow$ (both calls & puts!)
- $T \uparrow \Rightarrow C \uparrow$
- $r \uparrow \Rightarrow C \uparrow$

Put–call parity: $C - P = S_0 - Ke^{-rT}$. Calls and puts are two faces of the same coin.

Part III — The Greeks: sensitivities of the price

The Greeks are the partial derivatives of the option price with respect to each input. They tell you *how your P&L moves* when the world moves.

Greek	**Symbol**	**Measures sensitivity to**	**Definition**
Delta	Δ	underlying price S	$\partial C/\partial S$
Gamma	Γ	*change* in Delta	$\partial^2 C/\partial S^2$
Theta	Θ	passage of time (decay)	$\partial C/\partial t$
Vega	$\mathcal{V}$	volatility σ	$\partial C/\partial \sigma$
Rho	ρ	interest rate r	$\partial C/\partial r$

A position's total risk is the *sum* of its Greeks. Manage the book $\Rightarrow$ manage the Greeks.

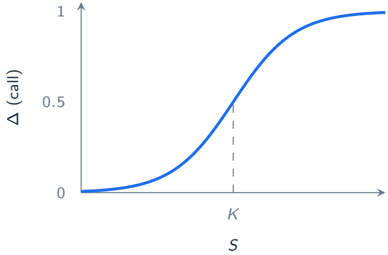
Delta: exposure to the underlying

$$\Delta = \frac{\partial C}{\partial S} = \mathrm{N}(d_1) \in [0,1] \text{ for a call,} \qquad \Delta_P = \mathrm{N}(d_1) - 1 \in [-1,0].$$

Three readings of Delta

- **Slope:** how much the option gains if S rises by \$1.
- **Equivalent shares:** a $\Delta = 0.6$ call behaves like holding 0.6 shares.
- **Risk-neutral probability** (roughly) that the option finishes in the money.

Deep ITM call $\Delta \to 1$; deep OTM $\Delta \to 0$; ATM $\Delta \approx 0.5$.



Delta runs smoothly from 0 to 1

Delta hedging: turning options into a risk-free recipe

To neutralise price risk, hold Δ shares *short* against a long call. The combined position is delta-neutral: to first order it does not move when S moves.

$$\Pi \;=\; C - \Delta\, S \qquad \Longrightarrow \qquad \frac{\partial \Pi}{\partial S} = 0.$$

- As S moves, Δ changes (that is Gamma), so the hedge must be rebalanced — this is **dynamic hedging**.
- Continuously rebalancing *replicates* the option: this is exactly the Black–Scholes argument in action.
- In practice you rebalance discretely; the mismatch is the cost/benefit of **Gamma**.

Market-makers quote options and then *delta-hedge* — they want the spread, not a directional bet.

Gamma, Theta, Vega — the second-order picture

Gamma Γ

- Curvature of value; how fast Δ moves.
- **Long** options $\Rightarrow$ long Gamma: you profit from big moves.
- Peaks **at the money**, near expiry.

Theta Θ

- Time decay: value lost per day, all else equal.
- Long options **pay** Theta; short options **earn** it.
- The price of holding convexity.

Vega $\mathcal{V}$

- Sensitivity to volatility σ.
- Long options are **long Vega**: gain if vol rises.
- Largest for longer-dated, ATM options.

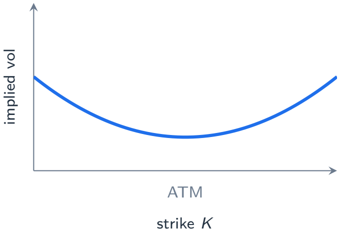
The fundamental trade-off: being long options gives you Gamma and Vega, but you *pay* Theta every day. Being short flips all three. There is no free convexity.

Implied volatility and the smile

Volatility is the one Black–Scholes input we cannot observe directly. So we invert the formula: the **implied volatility** is the σ that makes the model price equal the market price.

$$C_{\text{market}} = C_{\text{BS}}(\sigma_{\text{imp}}) \quad \Longrightarrow \quad \sigma_{\text{imp}}.$$

- IV is the market's *forward-looking* estimate of risk — an option's real "price" is quoted in vol.
- Empirically IV varies with strike: the volatility smile / skew.
- This shows the real world has fatter tails than the log-normal model assumes.

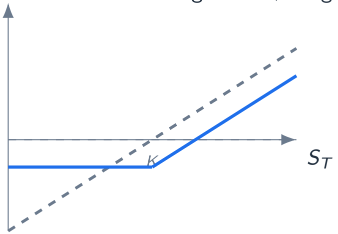


out-of-the-money options cost *more* vol

Part IV — Strategies I: shaping a stock position

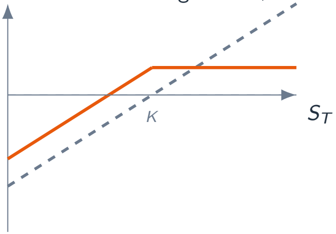
Options let you *reshape* the payoff of a holding.

Protective Put = long stock + long put



insures the downside; you pay the premium

Covered Call = long stock + short call



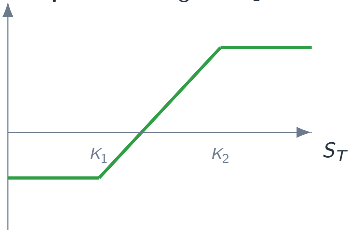
earns premium; caps the upside

Same underlying, two different risk profiles — you buy insurance, or you sell it.

Strategies II: spreads — a defined-risk directional bet

A vertical spread combines two options of the same type and expiry but different strikes. It caps *both* profit and loss and lowers the cost.

Bull Call Spread = long call K_1 + short call K_2

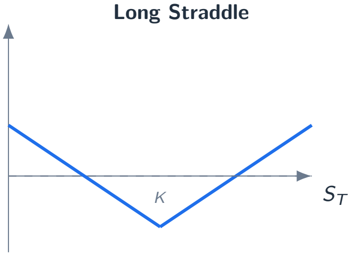


- **Max loss** = net premium paid (floor).
- **Max gain** = $(K_2 - K_1)-$ premium (ceiling).
- Cheaper than a bare call: you finance it by selling the upper strike.
- Mirror it with puts for a **bear put spread** — a defined-risk *down* bet.

Spreads express a *view with boundaries*: “up, but only this far, and I’ll risk only this much.”

Strategies III: trading volatility, not direction

A straddle (long call + long put, same strike) profits from a *big move either way* — it is a pure bet that realised volatility will beat the implied volatility you paid.



- **Long** straddle: profit if S moves far from K; loss if it sits still (you bleed Theta). Long Gamma & Vega.
- **Short** straddle: the opposite — earn premium if the market is calm, but open-ended risk if it jumps.
- A **strangle** uses OTM strikes: cheaper, needs a bigger move.

You can be right about *volatility* while having no view on *direction* at all.

Managing risk with the Greeks

A trading book holds many positions. Its total risk is the *aggregate* of Greeks — you manage the book by steering these numbers to target levels.

$$\Delta_{\text{book}} = \sum_i \Delta_i, \quad \Gamma_{\text{book}} = \sum_i \Gamma_i, \quad \mathcal{V}_{\text{book}} = \sum_i \mathcal{V}_i, \ \ldots$$

A typical workflow

1. Neutralise unwanted **Delta** with the underlying.
2. Cap **Gamma**/**Vega** with offsetting options.
3. Watch **Theta** — the daily cost of being long convexity.
4. Stress-test: "what if S jumps 10% and σ doubles?"

Broader tools

- **Value-at-Risk (VaR):** a loss threshold breached only with probability α.
- **Position limits** & **stop-losses**.
- **Diversification** across underlyings and expiries.
- Respect **tail risk** — models understate extremes.

Options are the most precise instrument we have for *sculpting* a risk profile — not

Key takeaways

The ideas

- Prices are **random**; volatility measures the risk.
- An **option** is a right, not an obligation — asymmetric payoff.
- Its value = **intrinsic** + **time value**, priced by **no-arbitrage** (Black–Scholes).
- The **Greeks** decompose that price into manageable risks.

The craft

- **Delta-hedge** to isolate the bet you actually want.
- Trade **direction** (spreads) or **volatility** (straddles) deliberately.
- Long options: pay **Theta** for **Gamma/Vega**. Short: the reverse.
- Manage the **book** by its aggregate Greeks and its tails.

Further reading: Hull, *Options, Futures & Other Derivatives* · Natenberg, *Option Volatility & Pricing*